

Modal Triplet Theory: Perturbative Coherent-Sector Quantum Gravity and the Heterotic UV-Completion Boundary

Peter Nero

July, 2026

Abstract

We separate the established low-energy quantum-gravity tier of Modal Triplet Theory (MTT) from its ultraviolet-completion problem. On a declared globally hyperbolic coherent background, the selected two-derivative Einstein–TEGR action admits the standard perturbative BV/BRST effective-field-theory construction, with gauge-invariant observables and the usual infrared graviton sector. We then re-examine the spectral proper-time (SPT) proposal. An exact graph theorem shows that Gaussian-damped lines control all loop directions if and only if their loop-momentum matrix has full rank; the mere presence of one damped graviton line is insufficient. We also prove that a nonzero propagator cannot simultaneously have a positive Källén–Lehmann/Stieltjes representation and permanent Gaussian decay in the same Euclidean spectral variable. Thus the earlier Stieltjes proof of unitarity for the Gaussian propagator is withdrawn, and SPT is retained only as a conditional filter/regulator model. The selected ultraviolet candidate is instead the q79 Fu–Yau heterotic branch. Conditional on an exact anomaly-free modular $(0, 2)$ worldsheet theory, GSO projection, factorization, and quantum-BV vertices, every fixed-genus, fixed-multiplicity amplitude has no local ultraviolet divergence. The holomorphic/HYM bundle, differential Bianchi representative, exact worldsheet packet, and all-genus or nonperturbative completion remain open.

Contents

1	Introduction	3
2	Claim tiers and the selected ultraviolet route	4
3	Standing assumptions and operator setting	5
3.1	Notation and constants (quick index)	6
4	Propagator and positivity	8
5	UV behaviour and one-loop modular check	8
5.1	Geometric origin of the UV regulator	8
6	Gravity–matter sector	10
7	Composite operators and BRST	10

8	Two-loop prototypes	11
9	Infrared behaviour and recovery of GR	11
10	Numerical illustrations	12
11	Derivation of SPT from projector structure	12
12	Proof of Theorem 1.1	13
A	Modular invariance at one loop	14
B	Laplace positivity and the Stieltjes compatibility no-go	14
C	Causality of retarded kernels	15
D	Gaussian domination of multi-loop graphs	16
E	BV/QME induction and anomalies	16
F	Derivation of SPT factorization	17
G	Microcausality of interacting fields	18
H	Master technical appendix: EG/pAQFT, BV–BRST & QME, FRG regulator removal, UV damping, unitarity	18
H.1	Standing hypotheses and inputs from the MTT spine	18
H.2	Conditional SPT damping after the support-gap axiom	18
H.3	Causal EG/pAQFT construction	19
H.4	BV data and the Classical Master Equation (CME)	19
H.5	All-orders QME and Ward/Slavnov–Taylor identities	19
H.6	Background-field FRG and regulator removal	19
H.7	Locality of counterterms and power counting in the graviton self-sector	20
H.8	Unitarity and microlocal spectrum	20
H.9	Optional all-line Euclidean convergence hypothesis	20
H.10	Projection vs. compactification (KK) in the coherent sector	20
H.11	Consolidated result	21
I	SPT Factorization and Gaussian UV Damping	21
I.1	Geometric and operator assumptions	21
I.2	Completely monotone filters and proper–time representation	21
I.3	SPT factorization	22
I.4	Kernel bounds and Gaussian decay	22
I.5	Filtered propagator and graphwise UV convergence	23
J	Derivation of the Proper-Time Support Gap	23
J.1	Purpose and logical role	23
J.2	Discrete schedule axiom	24
J.3	Construction of the effective filter	24
J.4	Complete monotonicity and Bernstein representation	24
J.5	Derivation of the proper-time support gap	25

J.6	SPT factorization	25
J.7	Local-momentum interpretation of Gaussian damping	25
J.8	Remarks on scope	26

1 Introduction

Perturbative quantum gravity in four dimensions, when naively quantized, is famously non-renormalizable. The current MTT construction closes the interacting low-energy Einstein–TEGR and quantum-GR EFT parity tier; it does not turn the four-dimensional Einstein action into a UV-complete theory. This paper identifies precisely what the proposed *Spectral Proper-Time* (SPT) filter does prove, corrects two overstrong implications in the previous version, and states the independent heterotic route selected by the q79 geometry.

Scope. All four-dimensional results are *perturbative* around a declared coherent MTT fixed point under bounded-geometry, hyperbolicity, Hadamard-state, and anomaly-cancellation hypotheses. “Background independence” is meant only in the BRST/background-field sense. Neither a preferred cosmic state nor a nonperturbative definition is inferred.

Theorem 1.1 (Main theorem: perturbative tier and ultraviolet boundary). *Let $(Y_4, \bar{g})$ be a globally hyperbolic bounded-geometry background, let the selected leading action be the Einstein–TEGR action coupled to anomaly-free matter, and choose a Hadamard state and admissible BV gauge fixing. If, in addition, an SPT filter is imposed, let C_Γ be the matrix sending independent loop momenta to the momenta of Gaussian-damped lines in a graph Γ . Then:*

1. **Perturbative EFT consistency.** *Epstein–Glaser/pAQFT renormalization and the standard BV induction construct the theory order by order. If the relevant ghost-number-one local cohomology is trivial, local counterterms restore the QME and physical observables are gauge independent. This is effective-field-theory renormalization, not ultraviolet finiteness.*
2. **Exact filtered-graph criterion.** *Polynomial vertices are Gaussian-dominated in every ultraviolet loop direction precisely when $\text{rank } C_\Gamma = L$, the number of independent loops. If the rank is smaller, the filter leaves an undamped ultraviolet direction and ordinary forest subtractions or power counting are still required.*
3. **Positivity–damping no-go.** *A nonzero positive Källén–Lehmann/Stieltjes propagator has a $1/x$ lower tail on some positive spectral interval and therefore cannot obey permanent Gaussian decay $O(e^{-\tau x}/x)$ for $\tau > 0$. The Gaussian model consequently needs an independent unitarity and causality construction.*
4. **Infrared recovery.** *At the selected two-derivative tier, the TT propagator and interactions are those of GR (equivalently TEGR up to its boundary term), with higher-derivative operators treated as EFT corrections.*
5. **Conditional heterotic inheritance.** *If the same q79 Fu–Yau background supplies an exact anomaly-free modular heterotic $(0, 2)$ SCFT, tachyon-free GSO projection, factorization, quantum-BV vertices, and an infrared/tadpole prescription, every fixed-genus fixed-multiplicity amplitude is free of local ultraviolet divergences. This coefficientwise statement does not prove convergence of the genus expansion or a nonperturbative theory.*

2 Claim tiers and the selected ultraviolet route

Closed low-energy tier. The current same-source chain selects the two-derivative Einstein–TEGR action, the two-helicity free graviton, and the interacting quantum-GR EFT observable functor at the same imported-parity standard used for ordinary perturbative gravity. This is a substantial closure result, but the nonzero two-loop Einstein counterterm remains part of that EFT and prevents its promotion to an all-scale theory.

SPT tier. The SPT filter is mathematically useful once a positive proper-time support gap is assumed. It is not a consequence of bounded projection alone, and the q79 source has not selected permanent Gaussian damping as its physical propagator. Sections B and D give the correct positivity no-go and graph-rank theorem. Accordingly, SPT is used below as a conditional filter or regulator analysis, not as the UV completion.

Selected heterotic tier. The primary UV route is the q79 rank-one Fu–Yau heterotic branch. The corpus constructs an explicit degree-two K3 incidence GLSM, the primitive rank-one torus source, the finite gerbe cocycle and its modular orbit data, and a smooth topological $SU(3)$ bundle candidate with $c_2 = 9u$ and $c_3 = \pm 6$. Fu–Yau existence theory and torsion linear sigma models supply the appropriate established mathematical framework [9, 1]. They do not automatically promote the topological candidate to the selected holomorphic/HYM bundle or exact worldsheet CFT.

Causal branch and double traversal. The physical-realization axiom, the retarded q79 representative, and a thermodynamic arrow are distinct data. A time-unoriented construction retains the conjugate pair q79/F and q369/F*; after a time orientation and retarded boundary problem are supplied, q79/F is selected as the representative. A closed-time-path double traversal then obeys the exact unitarity identity $Z[J, J] = 1$, but it neither derives the time orientation nor selects the initial state. The marked internal shared circle is common holonomy data and is not identified with compact physical time. The Schwinger–Keldysh formalism is the appropriate comparison framework for this doubled causal structure [10, 7].

Open UV-completion gates. The remaining same-source objects are the inverse-gerbe holomorphic spectral sheaf and local freeness, balanced HYM connection, differential Bianchi representative, full torsion-GLSM Fermi E/J data, seven analytic modular characters with GSO and factorization, and q79 string-field/BV vertices. Even after those are supplied, convergence of the genus expansion or a genuine nonperturbative definition is an independent final problem.

Version 5 correction record

Relative to v4, this version:

- withdraws the implication “positive heat kernel $\Rightarrow$ Stieltjes/Källén–Lehmann density” and gives an exact scalar counterexample;
- proves that positive spectral density is incompatible with permanent Gaussian ultraviolet decay for a nonzero propagator;
- replaces “one damped graviton line makes a graph finite” by the exact full-loop-rank criterion;

- reclassifies the proper-time gap as an explicit SPT hypothesis rather than a consequence of bounded projection;
- distinguishes perturbative EFT consistency, coefficientwise fixed-genus string UV finiteness, all-genus summability, and nonperturbative completion;
- adds the q79 heterotic, causal-branch, and closed-time-path boundaries.

3 Standing assumptions and operator setting

We assume throughout:

- bounded geometry on $(Y_4, \bar{g})$ and on the internal sector;
- a positive spectral gap $\lambda_* > 0$ above the coherent zero modes of A_{int} ;
- bounded projectors on Sobolev scales and a bounded metric shape map B ;
- commuting external/internal blocks $[E, A_{\text{int}}] = 0$.

SPT hypothesis (proper-time gap). In addition to the standing geometric and spectral assumptions above, the optional SPT model assumes a spectral filter with a strictly positive proper-time gap. Boundedness of a projector does not imply such a gap. Concretely, we assume the coherent projector can be realized (on the relevant slab) by completely monotone spectral filters on the external and internal blocks,

$$\Pi_{\text{coh}} = \phi(E) \psi(A_{\text{int}}), \quad \phi(z) = \int_{\tau_{\text{ext}}}^{\infty} e^{-tz} d\mu_{\text{ext}}(t), \quad \psi(\lambda) = \int_{\tau_{\text{int}}}^{\infty} e^{-s\lambda} d\mu_{\text{int}}(s),$$

with finite positive measures $\mu_{\text{ext}}, \mu_{\text{int}}$ and $\tau_{\text{ext}}, \tau_{\text{int}} > 0$. We write $\tau_0 := \min\{\tau_{\text{ext}}, \tau_{\text{int}}\} > 0$. This is the conditional SPT coarse-graining scale used in the filtered subsections; it is restated in Appendix A (Assumption A.3) and Appendix F. No primitive q79 selection of this support gap is claimed.

Stability under small off-diagonal couplings. If $A = E \oplus A_{\text{int}} + V$ with V symmetric and A_0 -form-bounded relative to $A_0 := E \oplus A_{\text{int}}$ with relative bound strictly less than 1, i.e. there exist $a \in [0, 1)$ and $b \geq 0$ such that

$$|\langle u, Vu \rangle| \leq a \langle u, A_0 u \rangle + b \|u\|^2 \quad \text{for all } u \in \text{Dom}(A_0^{1/2}),$$

then by Kato–Rellich (and Trotter product control) the heat semigroup satisfies, uniformly for $t \in (0, 1]$,

$$e^{-tA} = e^{-tE} e^{-tA_{\text{int}}} + \mathcal{O}(t) \quad \text{in operator norm on } H^1,$$

with constants depending continuously on (a, b) on bounded-geometry families. In particular, all SPT bounds persist with renormalised constants $(\tau_0, C_0) \mapsto (\tau'_0, C'_0)$. This includes mild warp/twist couplings in factorised backgrounds.

Lemma 3.1 (Proper-time representation for the projected propagator). *Assume the standing assumptions of Section 3 and, independently, the positive proper-time support-gap hypothesis above. Let $P = I \circ \Pi$ be the coherent projector and let A be the projected linearized graviton operator on the TT sector, with spectrum bounded below by $\lambda_* > 0$ on the Q -sector. Then there exists $\tau_0 > 0$ and constants $C, c > 0$ such that the projected propagator admits a proper-time representation with uniform bound*

$$\|\Delta_{\text{prop}}(k)\| \leq C \frac{e^{-\tau_0 k^2}}{k^2 + \lambda_*},$$

where k is the Fourier variable in locally geodesic coordinates on the slab and the bound is uniform on bounded-geometry charts.

Theorem 3.2 (SPT factorization under a support-gap hypothesis). *Under the above assumptions, including the positive support gap, there exists a bounded operator B_0 such that*

$$B = e^{-\frac{\tau_0}{2} E} B_0 e^{-\frac{\tau_0}{2} A_{\text{int}}},$$

with $\|B_0\| \leq C_0$. Consequently,

$$\|\Delta_{\text{prop}}(k)\| \leq \frac{C_0}{k^2 + \lambda_*} e^{-\tau_0 k^2},$$

so each internal graviton line carries a universal Gaussian factor $e^{-\tau_0 k^2}$.

Proof. By hypothesis, the coherent projector is realized by spectral filters on E and A_{int} whose representing measures have positive lower support. By bounded-geometry parametrix transfer, $DG(\Psi_*)e^{-t(-\Delta_{\text{prop}Y_4})} = e^{-ctE}S_t$ with $c \in (0, 1]$ and bounded S_t . Choosing $\tau_0 \leq \min\{c\tau_{\text{ext}}, \tau_{\text{int}}\}$ and factoring the semigroups yields the claimed decomposition. The proof consumes rather than derives the positive lower support. $\square$

Lemma 3.3 (Universality under projector realization). *Let $\Pi_{1,2}$ be two bounded realizations of the same coherent range and $B_i = DG(\Psi_*) \circ \Pi_i$. If $B_2 = B_1 U$ with U positive and commuting with E, A_{int} , then $\Delta_{\text{prop}1} = \Delta_{\text{prop}2}$. Physical correlators and the Gaussian bound are independent of projector filter details.*

3.1 Notation and constants (quick index)

Conventions. Background: $(Y_4, \bar{g})$ of bounded geometry; external operator E is the Lichnerowicz operator on TT fluctuations (de Donder/harmonic gauge); internal operator A_{int} acts on the orthogonal complement of the coherent sector. Blocks commute: $[E, A_{\text{int}}] = 0$. The metric shape map at the fixed point is B & bounded map & Metric shape map at Ψ^* ; $B = \frac{\delta g}{\delta \Psi}|_{\Psi^*}$; $\|B\| \leq C_B$; Sec. 2.

Symbol	Type	Meaning / Where fixed
$Y_4, \bar{g}$	geometry	4D background (bounded geometry); Sec. 3.
E	operator	Lichnerowicz operator on TT modes (external block).
A_{int}	operator	Internal block; spectral gap $\sigma(A_{\text{int}}) \subset [\lambda_*, \infty)$; Sec. 3.
A	operator	Full Hessian at the fixed point: $A = E \oplus A_{\text{int}}$; $[E, A_{\text{int}}] = 0$.
B	bounded map	Metric shape map at Ψ_* ; $\ B\ \leq C_B$; Sec. 3.
Π_{coh}	projector	Coherent-sector projector via spectral filters on $(-\bar{\square})$ and A_{int} .
B_0	bounded map	SPT core map after pulling out proper-time dressings; $\ B_0\ \leq C_0$; Thm. 3.2.
τ_0	time scale	SPT proper time; window $0 < \tau_0 \leq \min\{c\tau_{\text{ext}}, \tau_{\text{int}}\}$ (parametrix transfer $c \in (0, 1]$).
λ_*	gap	First positive eigenvalue bound for A_{int} on the non-coherent slice.
c_{proj}	constant	Optional benchmark normalization; not selected by the q79 source.
Λ^2	scale	Conditional filter scale $\Lambda^2 := c_{\text{proj}}\lambda_*$; Secs. 5, 10.
Δ_{prop}	kernel	Conditional filtered propagator $\Delta_{\text{prop}} = BA^{-1}B^*$; Secs. 4, 5.
K	operator	Quadratic TT kernel; the selected two-derivative limit is the GR/TEGR kernel.
F	function	Optional analytic filter normalized by $F(0) = 1$; Lem. 9.1.
ρ	measure	Hypothetical positive spectral measure used in the compatibility no-go; App. B.
Γ	action	Renormalized 1PI effective action in BV; Sec. 7.
$(\cdot, \cdot)$	bracket	BV antibracket; Sec. 7.
Δ_{BV}	operator	BV Laplacian (see note below to avoid clash with propagator Δ_{prop}).

Key inequalities at a glance. There exist constants $C_0, C_1 > 0$ (uniform on bounded-geometry families) such that

$$\textbf{SPT factorization: } B = e^{-\frac{\tau_0}{2}E} B_0 e^{-\frac{\tau_0}{2}A_{\text{int}}}, \quad \|B_0\| \leq C_0, \quad (3.1)$$

$$\textbf{Gaussian-dressed propagator: } \|\Delta_{\text{prop}}(k)\| \leq \frac{C_0}{k^2 + \lambda_*} e^{-\tau_0 k^2}, \quad (3.2)$$

$$\textbf{Full-loop-rank criterion: } \text{rank } C_\Gamma = L \iff C_\Gamma^T C_\Gamma > 0, \quad (3.3)$$

$$\textbf{IR expansion of } F: F(z) = 1 + C_{\text{IR}} z + \mathcal{O}(z^2), \quad z \rightarrow 0^+, \quad (3.4)$$

with $C_{\text{IR}} = F'(0)$ for the optional analytic filter.

Conventions (quick). Signature $(-, +, +, +)$; $\kappa_4^2 := 8\pi G$; background covariant derivative $\bar{\nabla}$; $\square_{\bar{g}} := \bar{g}^{\mu\nu} \bar{\nabla}_\mu \bar{\nabla}_\nu$. TT means transverse–traceless relative to $\bar{g}$. We use: Epstein–Glaser causal perturbation theory (EG), perturbative algebraic QFT (pAQFT), Functional Renormalization Group (FRG), Quantum Master Equation (QME), and BRST/BV in the background-field formalism. Throughout, Δ_{prop} denotes the graviton propagator and Δ_{BV} the BV Laplacian.

4 Propagator and positivity

Scope of positivity and unitarity. For the local Einstein–TEGR EFT, perturbative unitarity is asserted on the BRST/BV cohomology of observables in the EG/pAQFT framework, where gauge dependence and unphysical polarizations decouple. A Hadamard two-point function and time orientation are part of that construction.

Under the optional SPT assumptions, $\Delta_{\text{prop}} = BA^{-1}B^*$ has the positive heat-kernel form of Lemma B.1. That fact alone does not prove Osterwalder–Schrader positivity. Indeed, Theorem B.3 proves that the same nonzero propagator cannot have both a positive Källén–Lehmann density and permanent Gaussian decay in the same spectral variable. No unitarity claim for the physical Gaussian-dressed kernel is made in this version.

Relation to prior frameworks. The local theory uses the curved-spacetime EG/pAQFT locality and covariance programme. The Stieltjes and Osterwalder–Schrader literature [13, 15, 34, 25] instead supplies a diagnostic that the permanent Gaussian proposal fails, not its proof.

The Gaussian/entire damping below is a *conditional* SPT filter requiring the positive support-gap hypothesis. It is not derived from bounded projection alone.

5 UV behaviour and one-loop modular check

The Gaussian bound $\|\Delta_{\text{prop}}(k)\| \leq C_0(k^2 + \lambda_*)^{-1}e^{-\tau_0 k^2}$ ensures absolute convergence only when the damped-line momentum matrix has full loop rank, as proved in Theorem D.2. For a complete string background, modular invariance removes the point-particle short-proper-time region at one loop:

Proposition 5.1 (One-loop modular ultraviolet inheritance). *For an exact tachyon-free modular string partition function with the required GSO projection and level matching, integration over the modular fundamental domain removes the independent ultraviolet cusp $\tau_2 \rightarrow 0$. Finiteness of the full integral additionally requires control of the infrared degeneration $\tau_2 \rightarrow \infty$ and tadpoles.*

Proof sketch. Modular transformations identify the short-tube region with a representative inside the fundamental domain. They do not remove long-tube infrared degenerations. The q79 application is therefore conditional on the full worldsheet and infrared packet stated in Theorem 1.1. $\square$

5.1 Geometric origin of the UV regulator

Statement. Within the optional SPT support-gap hypothesis, a filtered coherent-sector model can be formed from (i) the bounded, joint harmonic projection $\Pi := \Pi_{B_1}\Pi_{B_2}\Pi_{B_3}$ on the internal bundles B_n , (ii) the internal pushforward I to the 4D base Y_4 , and (iii) a background-covariant coarse-graining on Y_4 implemented by a regulator $R_k(-\nabla^2)$ that is a positive function of the background Laplacian. The harmonic projection suppresses vertical modes; it does not by itself suppress arbitrarily large four-dimensional covectors. The FRG term is an auxiliary infrared coarse-graining device.

Bandlimiting by coherent projection. Let $A := \kappa_1\Delta_{B_1} + \kappa_2\Delta_{B_2} + \kappa_3\Delta_{B_3} + \varepsilon\Delta_Y$ on the 10D bundle (with $\kappa_n > 0$, $\varepsilon > 0$ fixed), and let $Q := \mathbf{1} - \Pi$ project to the non-coherent

(off-harmonic) sector. The internal gaps $\lambda_{n,*} > 0$ yield a uniform $\lambda_* := \min_n \kappa_n \lambda_{n,*} > 0$. By spectral calculus, for $t > 0$

$$\|A^{1/2}e^{-tA}Q\|_{L^2 \rightarrow L^2} \lesssim t^{-1/2}e^{-\lambda_*t}, \quad \|e^{-tA}\|_{L^2 \rightarrow H^1} \lesssim (1+t^{-1/2})e^{-\lambda_*t}. \quad (5.1)$$

Hence long-time modal flow suppresses Q -modes exponentially and contracts to $\text{Ran } \Pi$. The observable map $P := I \circ \Pi$ acts as a geometric band-limiter: external correlators carry only the fibre-harmonic content, with off-harmonic contributions damped by $e^{-\lambda_*t}$. This produces an internal spectral truncation, not a proof of four-dimensional ultraviolet finiteness.

Covariant regulator on Y_4 . On the observable side, define the (Euclidean) effective average action

$$\Gamma_k[g, \Phi] = S_{\text{coh}}[g, \Phi] + \frac{1}{2} \int \sqrt{g} \, \Phi R_k(-\nabla^2) \Phi, \quad (5.2)$$

with $R_k(z)$ satisfying $R_k(z) \rightarrow k^2$ for $z \ll k^2$ and $R_k(z) \rightarrow 0$ for $z \gg k^2$, and Φ denoting the collection of coherent fields (bosons, fermions in suitable quadratic forms). The flow is governed by the Wetterich equation

$$\partial_k \Gamma_k = \frac{1}{2} \text{STr} \left(\Gamma_k^{(2)} + R_k \right)^{-1} \partial_k R_k, \quad (5.3)$$

written in a background-field gauge so that R_k is a scalar function of $-\nabla^2$ and thus preserves diffeomorphism covariance of the flow. In Lorentzian signature one uses the standard in-in variant; all Ward/BRST identities are recovered as $k \rightarrow 0$.

With this background-covariant choice $R_k(-\bar{\nabla}^2)$, the modified Slavnov-Taylor identities and the QME hold at each finite k and reduce to the exact identities as $k \downarrow 0$ (see Theorem H.5).

SPT versus FRG (conceptual separation). It is important to distinguish the roles of the SPT factorization and the functional renormalization-group (FRG) regulator R_k . The SPT mechanism is a projector-induced filter considered after imposing the positive proper-time support gap. If retained at $k = 0$, it yields a Gaussian dressing $e^{-\tau_0 k^2}$ on selected lines, but its physical selection, unitarity, and causal support are not established. Its graphwise convergence scope is exactly the rank criterion of Theorem D.2.

By contrast, the FRG regulator $R_k(-\nabla^2)$ is introduced only as an organizational tool to define a covariant coarse-graining and to control the flow of local counterterms. The limit $k \rightarrow 0$ is taken after renormalization, and removing R_k does not remove the SPT Gaussian damping, which is fixed by the coherent-sector data (τ_0, λ_*) .

Propagator structure at fixed k . Because $P = I \circ \Pi$ projects to fibre-harmonic profiles, the 10D kinetic operator reduces to a 4D operator of the form $K_{\text{obs}}(-\nabla^2) + M_{\text{obs}}^2(x)$ on Y_4 . With the addition of $R_k(-\nabla^2)$ one has, schematically,

$$G_k(p^2) = \frac{1}{K_{\text{obs}}(p^2) + M_{\text{obs}}^2 + R_k(p^2)} \sim \begin{cases} 1/k^2 & (p^2 \ll k^2), \\ 1/K_{\text{obs}}(p^2) & (p^2 \gg k^2), \end{cases} \quad (5.4)$$

The internal bandlimiting from Π removes vertical high-frequency channels, while R_k regularizes low four-dimensional covectors in the flow equation. Since $R_k(p^2) \rightarrow 0$ at large p^2 , it is not an ultraviolet regulator and does not make bare loop integrals finite. A separate UV regularization and renormalization prescription is still required before taking $k \rightarrow 0$.

Compatibility with BV/QME. Since R_k is a function of the background Laplacian, the modified Slavnov–Taylor/BRST identities hold along the flow and reduce to the exact identities as $k \rightarrow 0$. The BV master equation is satisfied up to standard regulator insertions, which vanish in the physical limit; the proofs use only locality of the quadratic form and the spectral functional calculus for $R_k(-\nabla^2)$.

Conclusion. The geometry supplies an internal harmonic truncation, and $R_k(-\nabla^2)$ supplies a diffeomorphism-covariant FRG coarse-graining. Neither fact is a selected all-scale UV completion. The optional SPT filter gives additional graphwise damping only at the conditional tier stated above.

Remark. Appendix H.9 shows conditional Euclidean graph convergence if every internal line is assigned rapid entire damping. That stronger all-line hypothesis is not selected by current MTT geometry and is subject to the positivity and causality boundary of Theorem B.3.

6 Gravity–matter sector

At quadratic order around the fixed point, the Hessian is block–diagonal:

$$\mathcal{H} = \mathcal{H}_{\text{grav}} \oplus \mathcal{H}_{\text{matter}},$$

so the graviton propagator Δ_{prop} decouples from pure matter two–point functions. Mixed graviton–matter blocks vanish at the stationary background in de Donder/unitary gauges. Consequently, with SPT each graviton line carries the universal Gaussian damping, but a mixed diagram is Gaussian-dominated only when its damped-line matrix has full rank on every ultraviolet subgraph. Rank-deficient mixed subgraphs and pure matter subgraphs require the usual local renormalizations. These counterterms are precisely those already catalogued in the amplitudes layer (anomaly cancellations, hypercharge normalization, mod-2 SU(2) global check). See [18] for the full anomaly audit.

Scope of vertices. Throughout, interaction vertices are assumed to be local, background-covariant polynomials of finite tensorial rank in the fields and a finite number of background-covariant derivatives. This assumption is standard in pAQFT/EG renormalisation and permits Gaussian domination when the independent graph-rank hypothesis also holds (see App. D).

7 Composite operators and BRST

We quantize in the BV formalism with fields Φ , antifields Φ^* , and classical action S_0 solving $(S_0, S_0) = 0$. Denote the BV antibracket by $(\cdot, \cdot)$ and the BV Laplacian by Δ_{BV} . Let Γ be the renormalized effective action.

Theorem 7.1 (All-orders QME and Slavnov–Taylor identities). *Assume: (i) S_0 solves the CME $(S_0, S_0) = 0$; (ii) the 4D matter content is anomaly-free so that $H_{\text{loc}}^1(s|d) = 0$; and (iii) a local covariant renormalization prescription satisfying the quantum action principle is used. Then there exist local counterterms at every loop order such that*

$$\frac{1}{2}(\Gamma, \Gamma) = i\hbar \Delta_{\text{BV}}\Gamma.$$

as a formal perturbative identity to all loop orders.

Proof sketch. Proceed by induction on loop order L . Suppose the QME holds up to $L - 1$. At order L the breaking $\Delta_{\text{BV}}^{(L)}$ is a local integrated polynomial of ghost number $+1$. Consistency gives $s \Delta_{\text{BV}}^{(L)} = 0$, so $\Delta_{\text{BV}}^{(L)} \in H_{\text{loc}}^1(s|d)$. By anomaly freedom, this cohomology is trivial, hence $\Delta_{\text{BV}}^{(L)} = s \Xi^{(L)}$ for some local $\Xi^{(L)}$. Adding $S_{\text{ct}}^{(L)} = -\Xi^{(L)}$ restores the identity. Locality of the breaking follows from the causal renormalization construction, not from SPT convergence. $\square$

Proposition 7.2 (Nielsen identity (gauge-parameter independence)). *For gauge-fixing parameters ξ , there exists a local generator $\mathcal{N}(\xi)$ of a canonical BV transformation with*

$$\partial_\xi \Gamma = (\Gamma, \mathcal{N}(\xi)).$$

Thus BRST cohomology observables are ξ -independent.

Remark 7.3 (Unitarity and causality). For the local EFT, causal factorization, the Hadamard condition, and the ST identities give perturbative unitarity on BRST cohomology and microcausality. This remark does not apply to the permanent Gaussian kernel without the independent construction required in Section 4.

8 Two-loop prototypes

To illustrate Gaussian domination, consider the two-loop graviton-matter “sunset” self-energy.

Proposition 8.1 (Two-loop sunset bound). *Let $\Sigma_2(Q)$ be the two-loop graviton-matter self-energy with external momentum Q . Under Theorem 3.2, one has*

$$|\Sigma_2(Q)| \leq C \int d^4p d^4q \frac{e^{-\tau_0[p^2+q^2+(p+q+Q)^2]}}{(p^2 + \lambda_*)(q^2 + \lambda_*)((p+q+Q)^2 + \lambda_*)} \Pi(p, q, Q),$$

where Π is a polynomial fixed by vertex ranks. The integral converges absolutely and

$$|\Sigma_2(Q)| \leq C' P(Q) e^{-c \tau_0 Q^2},$$

for constants $C', c > 0$ and polynomial $P(Q)$.

Proof sketch. Each graviton propagator obeys $\|\Delta_{\text{prop}}(k)\| \leq C_0 e^{-\tau_0 k^2} / (k^2 + \lambda_*)$. Completing squares in (p, q) shows the Gaussian dominates the polynomial numerator and ensures absolute convergence. Exponential suppression in Q follows from the external factors after integration. $\square$

9 Infrared behaviour and recovery of GR

The projected kernel may be assigned a normalized form factor $F(\square)$ in the optional filtered model. We impose the normalization $F(0) = 1$ so that its low-energy expansion begins with the Einstein-Hilbert term. No positive Stieltjes representation is inferred for a permanently Gaussian F .

Lemma 9.1 (IR limit of a normalized analytic filter). *Let F be analytic in a neighborhood of zero, with $F(0) = 1$. Then*

$$F(0) = 1, \quad F(z) = 1 + C_{\text{IR}} z + \mathcal{O}(z^2), \quad z \rightarrow 0^+,$$

where $C_{\text{IR}} = F'(0)$. Thus a filter with variation scale Λ changes the GR propagator only by $\mathcal{O}(k^2/\Lambda^2)$ off shell at low covector norm.

Proof. This is the Taylor expansion of F at zero. $\square$

Remark 9.2 (Conditional static-potential illustration). For the particular Euclidean multiplier $F(k^2) = e^{-k^2/\Lambda^2}$, the static potential becomes $V(r) = -\frac{Gm}{r} \text{erf}(\Lambda r/2)$, which agrees with GR at $r \gg 1/\Lambda$ and is finite at $r = 0$. This is a property of that chosen filter, not a selected q79 prediction.

Remark 9.3 (Gravitational wave dispersion). If the entire multiplier has no zeros, it does not move the massless graviton pole and therefore does not by itself change the tree-level luminal dispersion relation. It changes off-shell amplitudes and short-distance Green functions.

10 Numerical illustrations

Numerical illustration, not a prediction. For the optional normalization used in v4, set

$$\Lambda^2 = c_{\text{proj}} \lambda_{\text{geo}}, \quad c_{\text{proj}} = \frac{1}{4}.$$

¹

Here λ_{geo} is a chosen positive geometric eigenvalue (in GeV^2). The value $c_{\text{proj}} = 1/4$ is a benchmark convention; it has not been selected by the current q79 source theorem.

As an anchor, take the Planck scenario with $\lambda_{\text{geo}} = 3.0 \times 10^{38} \text{GeV}^2$. Then

$$\Lambda = \frac{1}{2} \sqrt{\lambda_{\text{geo}}} \approx 8.66 \times 10^{18} \text{GeV},$$

so

$$1/\Lambda \approx 5.77 \times 10^{-20} \text{GeV}^{-1} \simeq 1.14 \times 10^{-35} \text{m}.$$

This agrees with the Planck suppression length because the Planck-sized λ_{geo} was supplied as the input. It is a consistency illustration, not a derivation of the Planck scale.

11 Derivation of SPT from projector structure

We show how the SPT factorization follows after adding the positive-support spectral-filter hypothesis to the bounded-projector realization. Let

$$\Pi_{\text{coh}} = \phi(-\Delta_{Y_4}^{\text{prop}}) \otimes \psi(A_{\text{int}}),$$

with ϕ, ψ given by proper-time integrals of the heat kernel.

Example (explicit $Y_4 \times T^d$ model). Let $X = Y_4 \times T^d$ with flat internal metric of radii L_i . Then $A_{\text{int}} = -\Delta_{T^d}$ has spectrum $\{4\pi^2 \sum_i n_i^2 / L_i^2\}$ with $\lambda_* = 4\pi^2 \min_i L_i^{-2}$. Let Π_{coh} project onto the joint zero-modes on T^d and low-frequency window on Y_4 given by a proper-time filter $\phi(-\square_{\bar{g}}) = \int_0^{\tau_{\text{ext}}} e^{-t(-\square_{\bar{g}})} \mu_{\text{ext}}(dt)$.

Convention remark. Here we use the complementary proper-time parametrization in which the coherent filter is written with support on $[0, \tau_{\text{ext}}]$ rather than $[\tau_0, \infty)$. The SPT proper-time gap corresponds in either convention to the exclusion of arbitrarily small proper times; the two parametrizations are equivalent under redefinition of the filter measure and lead to identical Gaussian suppression $e^{-\tau_0 k^2}$ on internal graviton lines.

¹The value $c_{\text{proj}} = 1/4$ is retained only to reproduce the v4 benchmark. The symmetric factorization does not by itself select a physical geometry-to-scale coefficient.

Example continued. Similarly take $\psi(A_{\text{int}}) = \int_0^{\tau_{\text{int}}} e^{-sA_{\text{int}}} \mu_{\text{int}}(ds)$ with positive measures. By Lemma F.1, $DG(\Psi^*)e^{-t(-\square_{\bar{g}})} = e^{-ctE} S_t$ with S_t bounded; choosing $\tau_0 \leq \min\{c\tau_{\text{ext}}, \tau_{\text{int}}\}$ yields

$$B = e^{-\frac{\tau_0}{2}E} B_0 e^{-\frac{\tau_0}{2}A_{\text{int}}}, \quad \|B_0\| \leq C_0.$$

Thus each filtered graviton line carries $e^{-\tau_0 k^2}$. For the optional numerical illustration we define $\Lambda^2 = c_{\text{proj}}\lambda_*$ and set $c_{\text{proj}} = 1/4$ as a benchmark convention.

Bounded-geometry parametrix transfer implies $DG(\Psi_*)e^{-t(-\Delta_{\text{BVY}_4})} = e^{-ctE} S_t$ with S_t bounded. Choosing $\tau_0 \leq \min\{c\tau_{\text{ext}}, \tau_{\text{int}}\}$ and factoring semigroups yields

$$B = e^{-\frac{\tau_0}{2}E} B_0 e^{-\frac{\tau_0}{2}A_{\text{int}}},$$

as in Theorem 3.2. This establishes the factorization inside the declared filter class; it does not select that class as the physical propagator.

12 Proof of Theorem 1.1

We prove the five tiered statements in Theorem 1.1. The local EFT and optional SPT filter are kept logically separate throughout.

Assumption 12.1 (Polynomial vertex growth). In the gauge-fixed TT sector, each interaction vertex factor is a polynomial (or rational function with bounded denominator away from IR singularities) in loop momenta, of total degree bounded by a constant depending only on the operator dimension of the interaction term. In particular, for each fixed loop order, the integrand is bounded by

$$|I(\ell)| \leq C(1 + \|\ell\|)^p \prod_j \|\Delta_j(\ell)\|,$$

for some $p < \infty$ and constant C depending on the graph but not on ℓ .

Step A: local perturbative theory. On a globally hyperbolic background with a Hadamard state, causal Epstein–Glaser splitting defines time-ordered products order by order. The standard BV consistency condition places a loop-order breaking in $H_{\text{loc}}^1(s \mid d)$. Under the anomaly hypothesis it is BRST exact and is removed by a local counterterm. This proves item (1) as an EFT statement.

Step B: optional filtered graphs. Under the independent support-gap hypothesis, Theorem 3.2 gives the line bound

$$\|\Delta_{\text{prop}}(k)\| \leq \frac{C_0}{k^2 + \lambda_*} e^{-\tau_0 k^2}. \quad (12.1)$$

Theorem D.2 diagonalizes the associated loop quadratic form $C_{\Gamma}^T C_{\Gamma}$. It is positive definite exactly at full column rank, proving item (2) and exposing the rank-deficient counterexample.

Step C: positivity boundary. Theorem B.3 supplies a positive-measure lower bound incompatible with exponential spectral decay. This proves item (3) and prevents the optional filtered result from being promoted by the old Stieltjes argument.

Step D: infrared law. The selected same-source two-derivative action is Einstein–TEGR. Standard gauge fixing therefore gives the GR TT propagator and interactions at this tier; higher-curvature terms remain EFT operators. This proves item (4) without requiring the Gaussian form factor.

Step E: heterotic inheritance. For an exact modular heterotic worldsheet, integration is over moduli-space fundamental domains rather than unrestricted point-particle Schwinger parameters. Interior ultraviolet collisions are treated by the worldsheet OPE and BRST factorization; boundary degenerations factorize onto lower-genus physical channels and are infrared/tadpole questions. Under the hypotheses listed in item (5), each fixed-genus coefficient therefore has no local UV divergence. The argument is coefficientwise and supplies neither convergence of the genus series nor a nonperturbative definition [27, 26, 28]. This proves item (5). $\square$

A Modular invariance at one loop

Theorem A.1 (Modular invariance). *The Narain partition function $Z_{\text{lat}}^{(k)}(\tau, \bar{\tau}; G, B)$ is invariant under $SL(2, \mathbb{Z})$ generated by $S : \tau \mapsto -1/\tau$, $T : \tau \mapsto \tau + 1$.*

Corollary A.2 (Weight-zero integrand). *Including ghosts and non-toroidal factors yields modular weight zero. Thus the one-loop graviton amplitude integral over the fundamental domain $\mathcal{F}$ has no independent short-tube ultraviolet cusp. Finiteness of the full integral additionally requires a tachyon-free GSO projection and control of the long-tube infrared degeneration.*

Displayed weight-zero check. Writing the one-loop integrand schematically as

$$\mathcal{I}(\tau, \bar{\tau}) = (\text{Im } \tau)^{-d/2} Z_{\text{lat}}^{(k)}(\tau, \bar{\tau}; G, B) Z_{\text{ghosts}}(\tau, \bar{\tau}) Z_{\text{rest}}(\tau, \bar{\tau}),$$

one has under $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL(2, \mathbb{Z})$:

$$Z_{\text{lat}}^{(k)}(\gamma \cdot \tau) = |c\tau + d|^{-d} Z_{\text{lat}}^{(k)}(\tau), \quad (\text{Im } \gamma \cdot \tau)^{-d/2} = |c\tau + d|^d (\text{Im } \tau)^{-d/2},$$

while Z_{ghosts} supplies weight -2 and Z_{rest} supplies $+2$, so the product has net weight 0. Thus $\mathcal{I}(\gamma \cdot \tau, \gamma \cdot \bar{\tau}) = \mathcal{I}(\tau, \bar{\tau})$.

Proof sketch. Poisson resummation exhibits Z_{lat} as an $O(d, d; \mathbb{Z})$ theta function; invariance under S, T follows. Adding ghost and measure factors balances modular weight. $\square$

B Laplace positivity and the Stieltjes compatibility no-go

Lemma B.1 (Conditional heat-kernel form). *Let $A = E \oplus A_{\text{int}}$ with commuting nonnegative blocks and let the bounded map B intertwine the external spectral calculus. Then*

$$\Delta_{\text{prop}} = BA^{-1}B^* = \int_0^\infty e^{-tE} M(t) dt, \quad M(t) = Be^{-tA_{\text{int}}} B^* \geq 0.$$

This is a positive Laplace representation. It does not, without an additional theorem, imply a Stieltjes representation in E .

Proof. The semigroup identity for A^{-1} and the commuting-block/intertwining hypotheses give the displayed integral. Positivity follows from $M(t) = N(t)^* N(t)$ with $N(t) = e^{-tA_{\text{int}}/2} B^*$. The class of Laplace transforms of positive kernels is the completely monotone class; Stieltjes functions form a proper subclass. $\square$

Proposition B.2 (The v4 Laplace-to-Stieltjes conversion is false). *Take the scalar example $M(t) = e^{-t}$ and $E = 1$. The heat integral is*

$$\int_0^\infty e^{-tE} M(t) dt = \frac{1}{2}.$$

The density proposed in v4 would be $\mu(s) = \int_0^\infty e^{-st} M(t) dt = (s+1)^{-1}$, but then

$$\int_0^\infty \frac{\mu(s)}{E+s} ds = \int_0^\infty \frac{ds}{(s+1)^2} = 1.$$

Hence that conversion does not represent the original propagator.

Theorem B.3 (Positive spectral density versus Gaussian decay). *Let*

$$D(x) = \int_0^\infty \frac{\rho(ds)}{x+s}, \quad x > 0,$$

for a nonzero positive measure ρ . Then D cannot obey $D(x) \leq Ce^{-\tau x}/(x+\lambda)$ for all sufficiently large x with $C, \tau, \lambda > 0$.

Proof. Choose $R < \infty$ with $m := \rho([0, R]) > 0$. Positivity gives $D(x) \geq m/(x+R)$. Combining the two bounds would imply

$$\frac{m}{C} \leq e^{-\tau x} \frac{x+R}{x+\lambda} \rightarrow 0,$$

a contradiction. $\square$

The theorem prevents the Gaussian SPT propagator from inheriting Osterwalder–Schrader positivity through the positive Källén–Lehmann argument used in v4 [20, 21]. It does not prove a general nonunitarity theorem for every nonlocal model; rather, unitarity and causality require a different construction. The local Einstein–TEGR EFT uses the standard Hadamard/BRST framework instead.

C Causality of retarded kernels

Proposition C.1 (Local EFT retarded support). *On a globally hyperbolic background, the gauge-fixed normally hyperbolic local Einstein–matter operator has unique retarded and advanced Green operators with causal support. Epstein–Glaser causal factorization preserves this support statement for renormalized local observables order by order.*

Proof sketch. This is the standard Green-hyperbolic and pAQFT construction for local operators. The time orientation and Hadamard state are inputs to the boundary-value problem. $\square$

Remark C.2 (Gaussian nonlocal kernel). The preceding proposition does not establish light-cone support for a permanently Gaussian-dressed nonlocal propagator. The v4 proof relied on the Stieltjes representation withdrawn in Theorem B.3. An independent causal operator or string-worldsheet construction is required before that filtered kernel can be treated as physical.

D Gaussian domination of multi-loop graphs

Lemma D.1 (Gaussian-dressed propagator). *Under Theorem 3.2, for any momentum $k \in \mathbb{R}^4$,*

$$\|\Delta_{\text{prop}}(k)\| \leq \frac{C_0}{k^2 + \lambda_*} e^{-\tau_0 k^2}.$$

Theorem D.2 (Damped-cycle rank criterion). *Let Γ have L independent loop momenta. Write the momentum of each Gaussian-damped line as*

$$k_j(\ell, Q) = \sum_{a=1}^L C_{ja} \ell_a + q_j(Q),$$

and let $C_\Gamma = (C_{ja})$. With polynomially bounded local vertices, the Gaussian factors suppress every ultraviolet loop direction if and only if

$$\text{rank } C_\Gamma = L.$$

When this full-rank condition holds, the corresponding Euclidean graph is absolutely ultraviolet convergent, apart from separately stated infrared singularities. When the rank is smaller, Gaussian damping alone gives no convergence conclusion in the null loop directions.

Compare with Weinberg's power counting for large momenta [31, 32];

Proof. The loop quadratic form is $C_\Gamma^T C_\Gamma$. If C_Γ has full column rank, its least singular value is positive. Completing the square in the Q -dependent shift gives a bound by a constant depending on Q times $\exp(-c_\Gamma \tau_0 |\ell|^2)$, which dominates every polynomial vertex factor. If the rank is smaller, choose $0 \neq v \in \ker C_\Gamma$. Along $\ell = \ell_0 + tv$ all damped line momenta are independent of t , so the Gaussian supplies no decay in that ultraviolet direction. Ordinary propagator power counting and forest renormalization then decide convergence. $\square$

Overlapping subgraphs. The full-rank test must hold for every ultraviolet subgraph after fixing its independent loop variables. Undamped matter subgraphs and rank-deficient mixed subgraphs require the ordinary forest/sector analysis. Fubini's theorem may be invoked directly only after absolute convergence has been established.

Vertex class used. Vertices are local, covariant polynomials in $h_{\mu\nu}$, matter fields, and a finite number of background-covariant derivatives (finite tensorial rank at each valence); the Gaussian bound is measured against this polynomial growth.

E BV/QME induction and anomalies

Local cohomology. We use $H_{\text{loc}}^1(s \mid d)$: the cohomology of local functionals at ghost number 1 modulo total derivatives, with differential given by the classical BRST operator s and the exterior derivative d . In $d = 4$ the pure diffeomorphism sector has *no* local anomaly, and with the anomaly-free matter content used here, $H_{\text{loc}}^1(s \mid d) = 0$ in the relevant sector.

Theorem E.1 (QME induction). *At loop order L , any breaking $\mathcal{A}^{(L)}$ of the QME lies in $H_{\text{loc}}^1(s \mid d)$. For anomaly-free matter content this cohomology is trivial, so $\mathcal{A}^{(L)} = s \Xi^{(L)}$. Adding $S_{\text{ct}}^{(L)} = -\Xi^{(L)}$ restores the QME. By induction, the QME holds to all orders.*

Proposition E.2 (Nielsen identity). *Gauge-parameter variation generates a canonical BV transformation: $\partial_\xi \Gamma = (\Gamma, \mathcal{N}(\xi))$. Physical observables are gauge independent.*

Remark E.3 (Anomaly audit). In 4D the SM matter content cancels all local gauge and mixed anomalies. Global SU(2) anomaly cancels (even number of doublets). Thus $H_{\text{loc}}^1(s|d) = 0$ in the model, justifying the induction.

F Derivation of SPT factorization

Lemma F.1 (Parametrix transfer and SPT factorisation: analytic setting). *Let $(Y^4, \bar{g})$ be globally hyperbolic with bounded geometry. Let E be the TT Lichnerowicz operator on $H_{\text{TT}}^1(Y)$ and $A_{\text{int}} \geq \lambda_* > 0$ on the incoherent complement, with $[E, A_{\text{int}}] = 0$. Assume the shape map B and the core map B_0 are bounded on Sobolev scales $H^s \rightarrow H^s$ ($s \in [0, 1]$). Then there exists $c \in (0, 1]$ and a bounded family S_t such that, as operators on H_{TT}^1 ,*

$$DG(\Psi^*) e^{-t(-\square_{\bar{g}})} = e^{-ctE} S_t, \quad \sup_{t \in (0, 1]} \|S_t\|_{H^1 \rightarrow H^1} < \infty,$$

and the Bochner integrals $\int_0^\infty e^{-tA} dt$ converge uniformly on bounded-geometry families. Choosing $\tau_0 \leq \min\{c\tau_{\text{ext}}, \tau_{\text{int}}\}$ and factoring the semigroups yields

$$B = e^{-\frac{\tau_0}{2}E} B_0 e^{-\frac{\tau_0}{2}A_{\text{int}}}, \quad \|B_0\| \leq C_0,$$

and the Gaussian bound $\|\Delta_{\text{prop}}(k)\| \leq C_0 e^{-\tau_0 k^2} / (k^2 + \lambda_)$.*

Proof. Bounded geometry gives standard heat-kernel parametrices with Gaussian bounds and Sobolev continuity; the transfer constant $c \in (0, 1]$ is the curvature/geometry loss in the parametrix conjugation. Spectral gap for A_{int} controls the vertical semigroup. Uniform Bochner convergence follows from the Gaussian bounds and the gap. Combine with Trotter product factorisation to obtain the displayed estimates. $\square$

Theorem F.2 (SPT factorization). *Let $\Pi_{\text{coh}} = \phi(-\Delta_{Y_4}) \otimes \psi(A_{\text{int}})$ with proper-time filters $\phi(z) = \int_{\tau_{\text{ext}}}^\infty e^{-tz} \mu_{\text{ext}}(dt)$ and $\psi(\lambda) = \int_{\tau_{\text{int}}}^\infty e^{-s\lambda} \mu_{\text{int}}(ds)$, where $\mu_{\text{ext/int}}$ are finite positive measures. On bounded-geometry backgrounds there exists $c \in (0, 1]$ and a family S_t bounded on Sobolev scales such that*

$$DG(\Psi_*) e^{-t(-\square_{\bar{g}})} = e^{-ctE} S_t \quad \text{for all } t \geq \tau_{\text{ext}}.$$

For any $\tau_0 \in (0, \min\{c\tau_{\text{ext}}, \tau_{\text{int}}\})$ define

$$B_0 := \int_{\tau_0/6}^\infty \int_{\tau_0}^\infty e^{-(ct-\tau_0)\frac{E}{2}} S_t DG(\Psi_*) e^{-(s-\tau_0)\frac{A_{\text{int}}}{2}} \mu_{\text{ext}}(dt) \mu_{\text{int}}(ds).$$

Then B_0 is bounded and

$$B = DG(\Psi_*) \Pi_{\text{coh}} = e^{-\frac{\tau_0}{2}E} B_0 e^{-\frac{\tau_0}{2}A_{\text{int}}}.$$

Proof. Insert the proper-time representations, use the transfer identity and factor out $e^{-\tau_0 E/2}$ and $e^{-\tau_0 A_{\text{int}}/2}$. The remaining Bochner integral defines B_0 ; boundedness follows from the finiteness of $\mu_{\text{ext/int}}$ and the uniform bounds on S_t . $\square$

G Microcausality of interacting fields

Proposition G.1 (EG construction and microlocal spectrum). *Let $\Delta_{\text{prop}_{\text{ret}}}$ be the retarded Green operator of the local gauge-fixed Einstein-matter kinetic operator on a globally hyperbolic background. With a Hadamard two-point function, time-ordered products constructed by Epstein-Glaser splitting are local and microcausal order by order after the required extensions at coincident points.*

This is the Epstein-Glaser construction [8], with microlocal control as in [6].

Proof. Green hyperbolicity gives retarded support, and the Hadamard microlocal spectrum condition controls products away from the total diagonal. The EG induction extends distributions to coincident configurations with local counterterms. This proof applies to the local EFT; it does not establish microcausality for the permanent Gaussian nonlocal kernel. $\square$

H Master technical appendix: EG/pAQFT, BV-BRST & QME, FRG regulator removal, UV damping, unitarity

H.1 Standing hypotheses and inputs from the MTT spine

We work on a globally hyperbolic Lorentzian 4-manifold $(Y^4, \bar{g})$ of bounded geometry on compact time slabs. The observable/coherent map is the bounded projector

$$P := I \circ \Pi : \Gamma(E_{10}) \rightarrow \Gamma(E_4),$$

where Π is the joint harmonic projector on the internal bundles and I pushes forward along the compact fibre; boundedness on L^2/H^1 and the closed, semibounded quadratic form for the reduced generator are standing hypotheses. The QFT layer chooses a Hadamard state and uses the curved-spacetime pAQFT/EG construction (causal factorisation, microlocal spectrum, local covariance). The GR layer provides background-field diffeomorphism covariance.

Remark H.1 (Scope and anomaly hypothesis). We assume the 4D diffeomorphism sector has no local anomaly (true in $d = 4$), and that any gauge matter is anomaly-free (as in your SM realisation). This is the only cohomological input in the QME proof.

H.2 Conditional SPT damping after the support-gap axiom

Proposition H.2 (Support-gap filter gives rapid spectral damping). *Assume the positive proper-time support gap and parametrix-transfer hypotheses of Theorem 3.2. Then the filtered two-point kernel can be written as*

$$\Delta_{\text{coh}} = f\left(-\bar{\nabla}^2 / \Lambda^2\right) \Delta_0, \quad \Lambda^2 \sim \tau_0^{-1} \sim \lambda_*,$$

for a heat-kernel filter f with rapid decay on the positive Euclidean spectral axis. In local momentum charts this gives the corresponding rapid covector decay. No reflection-positivity or causal-support conclusion follows.

Proof. Factor the lower endpoint τ_0 from the assumed proper-time measure and apply the heat-kernel covector estimate. The argument uses the support gap as an input and does not derive it from boundedness of Π . $\square$

H.3 Causal EG/pAQFT construction on $(Y^4, \bar{g})$

Theorem H.3 (Local, covariant time-ordered products on curved spacetime). *On globally hyperbolic $(Y^4, \bar{g})$ with a Hadamard two-point function, there exist multilinear maps $\mathcal{T}_n$ on local functionals satisfying: causal factorisation, locality & background covariance, microlocal spectrum (Hadamard wavefront sets), symmetry/adjointness, and controlled scaling degree. The renormalisation freedom at each order is a finite-dimensional affine space of local covariant counterterms.*

Proof. Epstein–Glaser splitting in curved spacetime using the microlocal spectrum condition; see [6, 11, 12, 24] Your QFT layer establishes the Hadamard selection and local covariance, so all hypotheses hold. $\square$

H.4 BV data and the Classical Master Equation (CME)

Let $\Phi = (h_{\mu\nu}, \text{matter}, \dots)$, C^μ the diffeo ghost, and Φ^* the antifields. The classical extended action $S_{\text{cl}}^{\text{BV}}$ (Einstein–Hilbert + matter + gauge-fixing + antifield couplings) satisfies:

Theorem H.4 (CME). $\{S_{\text{cl}}^{\text{BV}}, S_{\text{cl}}^{\text{BV}}\} = 0$.

Proof. Background-field representation of Diff is a Lie algebra; the induced BRST differential is nilpotent; equivalently the CME holds. Metric compatibility/Levi–Civita and Bianchi pushforward from your GR reduction ensure the background invariances used. $\square$

H.5 All-orders QME and Ward/Slavnov–Taylor identities

Let $S = S_{\text{cl}}^{\text{BV}} + \sum_{\ell \geq 1} \hbar^\ell S_\ell$ be the renormalised BV action and Δ_{BV} the BV Laplacian in the Hadamard regularisation. The QME reads:

$$\frac{i}{\hbar} \Delta_{\text{BV}} S + \frac{1}{2\hbar^2} \{S, S\} = 0.$$

Theorem H.5 (QME to all loop orders). *Assume (i) Theorem H.3; (ii) CME (Theorem H.4); (iii) trivial local BRST cohomology in ghost number 1 for diffeos and anomaly-free matter in $d = 4$. Then counterterms S_ℓ can be chosen so that the QME holds to all orders. Equivalently, background-field Ward/Slavnov–Taylor identities hold to all orders. [3, 4, 2, 22]*

Proof. Induction in $\hbar$. At order $\hbar^L$, the anomaly candidate $\mathcal{A}_L$ is a local functional of ghost number 1 (Quantum Action Principle). By (iii) the Wess–Zumino consistency implies $\mathcal{A}_L = s\mathcal{C}_L + d(\dots)$; choose $S_L := -i\mathcal{C}_L$ to cancel it. Locality and covariance are preserved by Theorem H.3. $\square$

H.6 Background-field FRG, modified Ward identities, and removal $k \rightarrow 0$

Add a background-covariant regulator $\Delta S_k = \frac{1}{2} \int \Phi R_k (-\bar{\nabla}^2) \Phi$; define Γ_k by Legendre transform; then

$$\partial_k \Gamma_k = \frac{i}{2} \text{STr}[(\Gamma_k^{(2)} + R_k)^{-1} \partial_k R_k].$$

At finite k , modified STI/QME hold with cutoff insertions.

Theorem H.6 (Regulator removal with STI/QME limit). *If R_k is background-covariant, positive, and vanishes as $k \downarrow 0$, then the renormalised correlators and Γ_k possess a $k \rightarrow 0$ limit, and the modified Ward/QME identities converge to the exact ones. The finite counterterms chosen in Theorem 7.1 can be implemented along the flow so that the limit solves the QME. [33, 5]*

Proof. Standard background-field FRG Ward analysis: the breaking terms are local and BRST-exact with coefficients $\propto \partial_k R_k$; they vanish as $k \rightarrow 0$. Heat-kernel/spectral estimates on $(Y, \bar{g})$ ensure the k -integral converges; locality/covariance is preserved. Compatibility with BV follows from the same cohomological argument as Theorem 7.1. $\square$

H.7 Locality of counterterms and power counting in the graviton self-sector

Theorem H.7 (Local, background-covariant renormalisation). *At each loop order the renormalisation freedom is a finite linear combination of diffeo-covariant local functionals built from $\sqrt{-\bar{g}}$, curvatures $(\bar{R}, \bar{R}_{\mu\nu}, \bar{R}_{\mu\nu\rho\sigma})$ and their covariant derivatives, together with BRST-exact gauge-fixing/ghost terms, consistent with the Ward identities.*

Proof. Directly from EG/pAQFT locality and covariance on globally hyperbolic spacetimes (Theorem H.3); background-field Ward identities restrict the tensors to diffeo-covariant combinations; BV cohomology fixes finite parts to preserve QME. $\square$

H.8 Unitarity and microlocal spectrum

Proposition H.8 (Bogoliubov S -matrix is unitary). *For Hermitian interactions the relative S -matrix constructed by EG/pAQFT is unitary order by order and satisfies causal factorisation.*

Proof. Adjointness and causal factorisation are axioms of the EG/pAQFT construction and hold on $(Y, \bar{g})$ with Hadamard states. Hence $S^\dagger S = 1$ order by order. $\square$

Proposition H.9 (Microlocal spectrum & composite operators). *Hadamard two-point structure and EG control imply that n -point functions satisfy the microlocal spectrum condition; point-splitting defines locally covariant composites (e.g. $\langle T_{\mu\nu} \rangle_{\text{ren}}$) with conservation inherited from background diffeomorphism covariance.*

H.9 Optional all-line Euclidean convergence hypothesis

Theorem H.10 (Conditional Euclidean convergence under all-line damping). *Assume that every internal Euclidean line carries a multiplier whose restriction to the positive spectral axis obeys $|f(x)| \leq C_N(1+x)^{-N}$ for every N . With polynomial local vertices, each fixed Feynman graph is absolutely ultraviolet convergent. This graph statement does not by itself prove reflection positivity, Lorentzian causality, the QME, or MTT selection of the multiplier.*

Proof. Choose N larger than the polynomial degree in every ultraviolet subgraph. The all-line decay then makes the Euclidean momentum integral absolutely convergent. The remaining physical properties are independent hypotheses. $\square$

H.10 Projection vs. compactification (KK) in the coherent sector

Proposition H.11 (Projection-KK equivalence (coherent sector)). *Let $\pi : \Sigma^{10} \rightarrow M^4$ be a Riemannian submersion with compact fibres F_6 , and Π the joint harmonic projector on F_6 . Under bounded geometry and a uniform fibre spectral gap, restricting fields to $\text{Ran } \Pi$ and pushing forward along π yields the same 4D effective action and Ward/Bianchi identities as KK truncation to fibre zero modes, up to operators suppressed by the fibre gap and warping.*

Proof. Zero-mode truncation coincides with restriction to $\text{Ran } \Pi$; pushforward of the 10D action produces the 4D Einstein–Hilbert term plus fibre curvature constants; the remainder is $O(\lambda_*^{-1}, \text{warp})$. Ward/Bianchi identities push forward because P commutes with base covariant derivatives and preserves the Noether charges. $\square$

H.11 Consolidated result

Theorem H.12 (Perturbative QG in the MTT coherent sector). *Under the hypotheses of §H.1, the local AQFT constructed on $(Y^4, \bar{g})$ by EG/pAQFT with BV–BRST data satisfies the QME and background-field Ward identities order by order after local renormalization, with locality, covariance, perturbative unitarity on BRST cohomology, and the microlocal spectrum condition. The optional SPT filter has only the graph-rank scope of Theorem D.2; no four-dimensional UV completion follows.*

I SPT Factorization and Gaussian UV Damping

This appendix derives the spectral proper-time (SPT) factorization from the explicit positive support-gap and transfer assumptions used in the filtered analysis. The support gap is an input; bounded geometry and bounded projection alone do not imply it.

I.1 Geometric and operator assumptions

Let (M, g) be a smooth Riemannian manifold of dimension d with bounded geometry: uniform bounds on curvature and its covariant derivatives and a positive injectivity radius. Let $E \rightarrow M$ be a Hermitian vector bundle with a compatible connection.

Let L be a nonnegative selfadjoint Laplace-type operator on $L^2(E)$, i.e.

$$L = \nabla^* \nabla + \mathcal{R},$$

where $\mathcal{R}$ is a smooth bundle endomorphism bounded below.

Assumption I.1 (Heat kernel bounds). The heat kernel $K_t(x, y)$ of e^{-tL} satisfies Gaussian upper bounds: there exist constants $C, c > 0$ such that for all $t > 0$,

$$\|K_t(x, y)\| \leq C t^{-d/2} \exp\left(-\frac{d(x, y)^2}{ct}\right).$$

Assumption I.1 holds for all Laplace-type operators on bounded-geometry manifolds.

I.2 Completely monotone filters and proper-time representation

Definition I.2 (Completely monotone filter). A function $f : [0, \infty) \rightarrow \mathbb{R}$ is *completely monotone* if

$$(-1)^m f^{(m)}(x) \geq 0 \quad \text{for all } x > 0, m \geq 0.$$

By Bernstein’s theorem, f is completely monotone if and only if there exists a finite positive Borel measure μ on $[0, \infty)$ such that

$$f(\lambda) = \int_0^\infty e^{-t\lambda} d\mu(t).$$

Assumption I.3 (Proper-time gap). The representing measure μ of f satisfies

$$\text{supp}(\mu) \subset [\tau_0, \infty) \quad \text{for some } \tau_0 > 0.$$

Replacement of Assumption A.3 (Proper-Time Support Gap). The discrete schedule used here is an additional model axiom: every evolve–project cycle $T_\tau = \Pi_{\text{coh}} \circ \Phi_\tau$ is assumed to apply a smoothing semigroup for a strictly positive minimum duration $\tau_* > 0$. Given that axiom, Appendix J proves that the filter measure is supported in $[\tau_*, \infty)$ and yields the SPT factorization. The argument does not derive the minimum duration from the abstract fixed-point or q79 geometry.

Definition I.4 (Spectral filter). Define the bounded operator

$$B := f(L)$$

by functional calculus.

Lemma I.5 (Exact proper–time representation). *Under Assumption I.3,*

$$B = \int_{\tau_0}^{\infty} e^{-tL} d\mu(t),$$

where the integral converges in the strong operator topology (and in operator norm if μ has finite total mass).

Proof. By Bernstein’s theorem and Assumption I.3,

$$f(L) = \int_0^{\infty} e^{-tL} d\mu(t) = \int_{\tau_0}^{\infty} e^{-tL} d\mu(t).$$

Strong convergence follows from monotone convergence and the contractivity of e^{-tL} on L^2 . $\square$

I.3 SPT factorization

Lemma I.6 (Spectral proper–time factorization). *Define*

$$B_0 := \int_0^{\infty} e^{-sL} d\mu(\tau_0 + s).$$

Then

$$B = e^{-\frac{\tau_0}{2}L} B_0 e^{-\frac{\tau_0}{2}L},$$

and B_0 is a bounded positive operator commuting with L .

Proof. By Lemma I.5, substitute $t = \tau_0 + s$:

$$B = \int_{\tau_0}^{\infty} e^{-tL} d\mu(t) = e^{-\frac{\tau_0}{2}L} \left(\int_0^{\infty} e^{-sL} d\mu(\tau_0 + s) \right) e^{-\frac{\tau_0}{2}L}.$$

Positivity and commutation with L follow from functional calculus. $\square$

I.4 Kernel bounds and Gaussian decay

Let $K_B(x, y)$ denote the Schwartz kernel of B .

Lemma I.7 (Kernel bound). *Under Assumption I.1,*

$$\|K_B(x, y)\| \leq C \int_{\tau_0}^{\infty} t^{-d/2} \exp\left(-\frac{d(x, y)^2}{ct}\right) d\mu(t).$$

Proof. Insert the heat kernel bound of Assumption I.1 into the representation of Lemma I.5 and integrate against $d\mu(t)$. $\square$

Lemma I.8 (Gaussian Fourier multiplier bound). *In a bounded-geometry coordinate chart, the Fourier transform of K_B satisfies*

$$\|\widehat{K_B}(\xi)\| \leq C' e^{-\tau_0|\xi|^2},$$

uniformly for $|\xi|$ sufficiently large.

Proof. In local coordinates, the Fourier transform of e^{-tL} is bounded by $\exp(-t|\xi|^2)$ up to polynomial prefactors. Since $t \geq \tau_0$ on the support of μ , the integral representation yields

$$\|\widehat{K_B}(\xi)\| \leq \int_{\tau_0}^{\infty} C'' e^{-t|\xi|^2} d\mu(t) \leq C' e^{-\tau_0|\xi|^2}.$$

$\square$

I.5 Filtered propagator and graphwise UV convergence

Let L be the projected graviton operator in the TT sector and let

$$\Delta := (L + m^2)^{-1}.$$

Define the filtered propagator

$$\Delta_{\text{filt}} := B \Delta.$$

Proposition I.9 (Gaussian UV damping of the propagator). *There exist constants $C, \lambda_* > 0$ such that in local momentum variables,*

$$\|\Delta_{\text{filt}}(k)\| \leq C \frac{e^{-\tau_0 k^2}}{k^2 + \lambda_*}.$$

Proof. Combine the resolvent bound $(k^2 + \lambda_*)^{-1}$ with the Fourier multiplier estimate of Lemma I.8. $\square$

Corollary I.10 (Full-rank filtered-graph convergence). *Assume that each interaction vertex grows at most polynomially in momenta and that the filtered-line momentum matrix has rank L on the graph and each UV subgraph, where L is the corresponding loop number. Then the graph is absolutely convergent in all ultraviolet loop directions.*

Proof. This is Theorem D.2 applied to the filtered propagator bound. $\square$

Remark I.11. Pure matter subgraphs without internal graviton lines renormalize by standard EFT methods and are not claimed UV finite by this mechanism.

J Derivation of the Proper-Time Support Gap

J.1 Purpose and logical role

In the constructive quantum-gravity analysis, several results rely on the existence of a strictly positive proper-time support gap

$$\text{supp}(\mu) \subset [\tau_0, \infty), \quad \tau_0 > 0, \tag{J.1}$$

for the Bernstein measure μ associated with the filter defining the coherent projector. In earlier sections this condition appeared as an explicit assumption (denoted “Assumption A.3”).

The purpose of this appendix is to *derive* (J.1) from a minimal and physically interpretable dynamical axiom: the existence of a strictly positive minimum smoothing time per evolve–project cycle. This removes any ambiguity about whether the proper-time gap is imposed ad hoc or follows from the operational structure of the theory.

Throughout this appendix we work under the bounded-geometry and Laplace-type assumptions already in force in the main text.

J.2 Discrete schedule axiom

Axiom 1 (Discrete projection schedule). Physical evolution proceeds in discrete evolve–project cycles. Each cycle applies a smoothing semigroup $S_t = e^{-tL}$ with a strictly positive minimum duration $\tau_* > 0$. No admissible physical update applies smoothing for times $t < \tau_*$.

This axiom formalizes the idea that admissible dynamics has a finite coarse-graining cadence and excludes arbitrarily short proper-time updates. It is independent of spectral gap assumptions and does not refer to any particular representation of the projector.

J.3 Construction of the effective filter

Let $L \geq 0$ be a self-adjoint Laplace-type operator acting on a Hilbert space (or section space over a bounded-geometry slab). Define the heat semigroup $S_t = e^{-tL}$.

Consider an effective filter obtained by iterating the evolve–project cycle a random number of times, with a geometric distribution of cycle counts:

$$F = \sum_{n=1}^{\infty} p(1-p)^{n-1} S_{n\tau_*}, \quad p \in (0, 1). \quad (\text{J.2})$$

This choice is technically convenient and physically natural: each update applies at least one full cycle of duration τ_* , and the distribution has finite mean.

J.4 Complete monotonicity and Bernstein representation

Lemma J.1 (Complete monotonicity). *The scalar function*

$$f(\lambda) = \sum_{n=1}^{\infty} p(1-p)^{n-1} e^{-n\tau_*\lambda}, \quad \lambda \geq 0, \quad (\text{J.3})$$

is completely monotone on $[0, \infty)$.

Proof. Each term $e^{-n\tau_*\lambda}$ is completely monotone, and the series converges absolutely and uniformly on compact subsets of $[0, \infty)$ since $(1-p)e^{-n\tau_*\lambda} \leq 1-p < 1$. Complete monotonicity is preserved under positive linear combinations. $\square$

By Bernstein’s theorem, f admits a unique representation as a Laplace transform of a positive measure.

Lemma J.2 (Explicit Bernstein measure). *The Bernstein measure associated with f is*

$$\mu = \sum_{n=1}^{\infty} p(1-p)^{n-1} \delta_{n\tau_*}. \quad (\text{J.4})$$

Proof. Each exponential $e^{-n\tau_*\lambda}$ is the Laplace transform of $\delta_{n\tau_*}$. Linearity of the Laplace transform yields (J.4). $\square$

J.5 Derivation of the proper-time support gap

Theorem J.3 (Proper-time support gap). *Under Axiom 1, the Bernstein measure μ of the effective filter $F = f(L)$ satisfies*

$$\text{supp}(\mu) \subset [\tau_*, \infty). \quad (\text{J.5})$$

In particular, the proper-time support gap exists with $\tau_0 = \tau_$.*

Proof. From Lemma J.2, the support of μ consists of the discrete set $\{n\tau_* \mid n \in \mathbb{N}\}$. The smallest element is τ_* , hence the support is contained in $[\tau_*, \infty)$. $\square$

This theorem replaces the previously stated proper-time gap assumption. The gap is no longer an independent hypothesis but a direct consequence of the discrete update cadence.

J.6 SPT factorization

Corollary J.4 (Spectral proper-time factorization). *Let $F = f(L)$ be defined as above. Then*

$$F = e^{-\frac{\tau_*}{2}L} F_0 e^{-\frac{\tau_*}{2}L}, \quad (\text{J.6})$$

where

$$F_0 := \int_{\tau_*}^{\infty} e^{-(t-\tau_*)L} d\mu(t) \quad (\text{J.7})$$

is a bounded, positive operator commuting with L .

Proof. This follows immediately by inserting the Bernstein representation and factoring out $e^{-\frac{\tau_*}{2}L}$ on both sides. Boundedness and positivity of F_0 follow from positivity of μ and standard semigroup bounds. $\square$

J.7 Local-momentum interpretation of Gaussian damping

We emphasize that $e^{-\tau_*L}$ yields Gaussian suppression in the *spectral* variable of L . On curved backgrounds there is no global Fourier momentum. The correct interpretation is microlocal.

Proposition J.5 (Local UV Gaussian damping). *Let L be a Laplace-type operator on a bounded-geometry slab. Then the principal symbol of $e^{-\tau_*L}$ is*

$$\sigma_{\text{pr}}\left(e^{-\tau_*L}\right)(x, \xi) = e^{-\tau_*\sigma_{\text{pr}}(L)(x, \xi)} = e^{-\tau_*|\xi|_g^2}, \quad (\text{J.8})$$

where $|\xi|_g^2$ is the local covector norm induced by the metric.

Proof. For Laplace-type operators, $\sigma_{\text{pr}}(L)(x, \xi) = |\xi|_g^2$. Since the function $\lambda \mapsto e^{-\tau_*\lambda}$ is entire, the standard pseudodifferential functional calculus applies, yielding that the principal symbol of $e^{-\tau_*L}$ is obtained by functional composition with $\sigma_{\text{pr}}(L)$. $\square$

Thus, statements such as “Gaussian damping $e^{-\tau_*k^2}$ ” should be read as shorthand for Gaussian damping in the local UV covector variable $|\xi|_g^2$. In local inertial or asymptotically flat limits, this reduces to the familiar momentum-space expression.

J.8 Remarks on scope

- The derivation above consumes the discrete schedule/support-gap axiom. Current q79 geometry has not promoted that axiom to a selected physical law.
- The Gaussian damping statement is local and microlocal; no claim of a global Minkowski momentum variable is made on arbitrary curved backgrounds.
- Extensions to specific sectors (e.g. TT gravitons) require only that the corresponding kinetic operator be Laplace-type up to lower-order curvature terms, as assumed elsewhere in the paper.

This completes the conditional factorization from the stated support-gap axiom.

References

- [1] Allan Adams, Morten Ernebjerg, and Joshua M. Lapan. Linear models for flux vacua. *Advances in Theoretical and Mathematical Physics*, 12:817–852, 2008.
- [2] Glenn Barnich, Friedemann Brandt, and Marc Henneaux. Local brst cohomology in gauge theories. *Physics Reports*, 338(5):439–569, 2000.
- [3] Igor A. Batalin and Grigori A. Vilkovisky. Gauge algebra and quantization. *Physics Letters B*, 102(1):27–31, 1981.
- [4] Igor A. Batalin and Grigori A. Vilkovisky. Quantization of gauge theories with linearly dependent generators. *Physical Review D*, 28:2567–2582, 1983.
- [5] Jürgen Berges, Nikolaos Tetradis, and Christof Wetterich. Non-perturbative renormalization flow in quantum field theory and statistical physics. *Physics Reports*, 363(4-6):223–386, 2002.
- [6] R. Brunetti and K. Fredenhagen. Microlocal analysis and interacting quantum field theories: Renormalization on curved spacetimes. *Communications in Mathematical Physics*, 208:623–661, 2000.
- [7] Michael Crossley, Paolo Glorioso, and Hong Liu. Effective field theory of dissipative fluids. *Journal of High Energy Physics*, 2017(9):095, 2017.
- [8] Henri Epstein and V. Glaser. The role of locality in perturbation theory. *Annales de l’Institut Henri Poincaré A*, 19:211–295, 1973.
- [9] Ji-Xiang Fu and Shing-Tung Yau. The theory of superstring with flux on non-kähler manifolds and the complex monge–ampere equation. *Journal of Differential Geometry*, 78(3):369–428, 2008.
- [10] Felix M. Haehl, R. Loganayagam, and Mukund Rangamani. Schwinger–keldysh formalism i: Brst symmetries and superspace. *Journal of High Energy Physics*, 2017(6):069, 2017.
- [11] Stefan Hollands and Robert M. Wald. Local wick polynomials and time ordered products of quantum fields in curved spacetime. *Communications in Mathematical Physics*, 223:289–326, 2001.

- [12] Stefan Hollands and Robert M. Wald. Existence of local covariant time ordered products of quantum fields in curved spacetime. *Communications in Mathematical Physics*, 231:309–345, 2002.
- [13] Gunnar Källén. On the definition of the renormalization constants for quantized fields. *Helvetica Physica Acta*, 25:417–434, 1952.
- [14] Tosio Kato. *Perturbation Theory for Linear Operators*. Springer, classics in mathematics edition, 1995.
- [15] Harry Lehmann. On the properties of propagation functions and renormalization constants of quantized fields. *Il Nuovo Cimento*, 11:342–357, 1954.
- [16] K. S. Narain. New heterotic string theories in uncompactified dimensions < 10 . *Physics Letters B*, 169(1):41–46, 1986.
- [17] K. S. Narain, M. H. Sarmadi, and Edward Witten. A note on toroidal compactification of heterotic string theory. *Nuclear Physics B*, 279(3–4):369–379, 1987.
- [18] Peter Nero. Modal triplet theory: Quantum amplitudes from modal geometry, 2025. Community: Modal Triplet Theory (MTT).
- [19] N. K. Nielsen. On the gauge dependence of spontaneous symmetry breaking in gauge theories. *Nuclear Physics B*, 101:173–188, 1975.
- [20] K. Osterwalder and R. Schrader. Axioms for euclidean green’s functions. *Communications in Mathematical Physics*, 31:83–112, 1973.
- [21] K. Osterwalder and R. Schrader. Axioms for euclidean green’s functions ii. *Communications in Mathematical Physics*, 42:281–305, 1975.
- [22] Olivier Piguet and Silvio P. Sorella. *Algebraic Renormalization: Perturbative Renormalization, Symmetries and Anomalies*, volume 28 of *Lecture Notes in Physics Monographs*. Springer, 1995.
- [23] Joseph Polchinski. *String Theory, Vol. 1: An Introduction to the Bosonic String*. Cambridge University Press, 1998.
- [24] Kasia Rejzner. *Perturbative Algebraic Quantum Field Theory: An Introduction for Mathematicians*. Mathematical Physics Studies. Springer, 2016.
- [25] René L. Schilling, Renming Song, and Zoran Vondraček. *Bernstein Functions: Theory and Applications*. De Gruyter, 2nd edition, 2012.
- [26] Ashoke Sen. Superstring perturbation theory revisited. *Journal of High Energy Physics*, 2015(11):008, 2015.
- [27] Ashoke Sen. Ultraviolet and infrared divergences in superstring theory, 2015. KIAS Newsletter invited article.
- [28] Ashoke Sen. Bv master action for heterotic and type ii string field theories. *Journal of High Energy Physics*, 2016(2):087, 2016.
- [29] R. F. Streater and A. S. Wightman. *PCT, Spin and Statistics, and All That*. Benjamin, 1964.

- [30] H. F. Trotter. On the product of semi-groups of operators. *Proceedings of the American Mathematical Society*, 10:545–551, 1959.
- [31] Steven Weinberg. High-energy behavior in quantum field theory. *Physical Review*, 138:B988–B1002, 1965.
- [32] Steven Weinberg. *The Quantum Theory of Fields, Vol. 1*. Cambridge University Press, 1995.
- [33] Christof Wetterich. Exact evolution equation for the effective potential. *Physics Letters B*, 301(1):90–94, 1993.
- [34] David V. Widder. *The Laplace Transform*. Princeton University Press, 1941.